Worksheet: Taxonomy of Data

Names:

1. What type of variable is being measured by each of the following questions?

A.

How much coding experience do you have?

- None.
- Very little. I've just dipped my toe in.
- A bit. I've used coding in a limited capacity, such as for a small part of a course
- Some. I've had to write code for one or two classes and am comfortable with it.
- A good deal. I have several years of experience writing code.

E.

What year are you in your studies?

- 1st year / freshman
- 2nd year / sophomore
- 3rd year / junior
- 4th+ year / senior

C.

Please also rate that experience on a scale from 1 to 10.

	1	2	3	4	5	6	7	8	9	10	
No experience	○	○	○	○	○	○	○	○	○	○	Tons of experience

F.

Are hotdogs sandwiches?

- Yes
- No

D.

What zip code do you consider home? (If from abroad, use UC Berkeley's 94720)

G.

What is the probability that a new COVID variant will disrupt instruction this semester?

2. Last class you collected data on six different variables on your classmates. List the data types of each of those variables, the units that they were measured in (if it's continuous numerical), and the range of possible values that you would expect them to take.

- Name:
- Contact info:
- Hometown:
- # Siblings:
- # Semesters in college:
- Farthest distance from campus during the break:

3. As discussed in the lecture notes, this is **not** a data frame:

		Alien color		
		Green	**Blue**	**Total**
Time of sighting	**Night**	55	11	**66**
	Day	23	55	**78**
	Total	**78**	**66**	**144**

Think about **where this summary table came from**. Imagine you are researchers who are collecting this data, and you record your observations as rows in a dataframe.

- What would the *unit* of observation be?

- How many rows would there be? How many columns would there be?

- Sketch a schematic of the data frame based off of your answers to the above (just column names and a few example rows)

4. Pull out your handout from yesterday's class again! You'll now be storing some of the data you collected using R code.

- Open R Studio (website > top right area > (R) button)
- Create a new .qmd file, name it `taxonomy-of-data`, and **save** it.
- Insert a new code cell.
- Create three vectors: `name`, `hometown`, and `sibs_and_pets` that contain observations on those variables from 6 people in this class. Copy the code you used below.
- **You don't need to turn in the code file itself** - just copy down the code you wrote therem once it's working,

- Combine the three vectors into a data frame called `my_classmates`. Copy the code you used below.

Didn't read today's tutorial? It's an important one, and will really help solidify your coding foundation that you'll use for the rest of the course.

Want more practice with data types and data frames? On the main website, this week under "extras," you'll find a Data Types and Data Frames link with tons more practice problems like this. We suggest you try a few now, and use the others as study tools before the quiz.